Mahmoud Saeed Mansour

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Education

Pharos University 2022 – 2026
Bachelor of Science in Computer Science & Artificial Intelligence, GPA:3.76 *Alexandria , Egypt*

Experience

iSchool July 2026 – Aug 2026
Tutor *Remote, Egypt*

- Delivered interactive sessions in Python, AI, Data Science, and 2D Game Design (Godot) for students aged 6–18.
- Guided students in building, debugging, and presenting programming projects.

Projects

Image Viewer with Filters and Classification | *Python, PyQt5, OpenCV, TensorFlow, CNN, Image Processing*

- Developed a PyQt5 desktop application for image loading, visualization, and processing.
- Implemented image processing techniques including Canny, Median filtering, Otsu thresholding, Harris corners, and SIFT.
- Integrated a CNN model to classify images into six scene categories.
- Enabled side-by-side visualization of original and processed images with a pre-trained model for immediate use.

Tic Tac Toe Game | *Dart, Flutter*

- Developed a mobile application using the Flutter framework with two play modes: against the bot or against a friend. Implemented an AI algorithm for the bot and integrated win/lose detection with options to replay or exit the game.

Face Mask Segmentation during COVID-19 | *Python, CNN, TensorFlow, Keras, Flask, HTML, CSS*

- Built a binary semantic segmentation model for face mask detection using ResNet50 and VGG16 with custom upsampling layers for accurate segmentation.
- Trained the model on dataset for mask and deployed it as a Flask web application with a custom HTML/CSS interface for real-time mask detection.

Real-time Sign Language Recognition | *Python, OpenCV, MediaPipe, Flask, scikit-learn*

- Developed a real-time ASL recognition system using MediaPipe Hands and OpenCV.
- Extracted hand landmarks and trained a Random Forest model to classify 14 ASL signs.
- Deployed the model as a Flask web application with live webcam recognition.

Real-time Object Detection | *Python, YOLOv8, OpenCV, Tkinter*

- Developed a YOLOv8-based object detection system for self-driving applications with support for webcam and video input.
- Created a Tkinter GUI with FPS display, bounding boxes, class labels, and efficient threaded video streaming for real-time performance.

Real-time Lane Detection | *Python, ResNet50, Kalman Filter, Hough Transform, OpenCV, Tkinter*

- Implemented multiple approaches for lane detection: segmentation with ResNet50, Kalman filter tracking, and Hough transform.
- Developed a Tkinter GUI with OpenCV integration for real-time lane detection, FPS monitoring, and annotated video streaming.

Real-time Traffic Light Detection | *Python, YOLOv8, PyTorch, OpenCV*

- Trained a custom YOLOv8 model on a traffic light dataset to classify signals as Green, Yellow, or Red. Saved the model in ONNX format for cross-platform deployment.
- Implemented real-time detection with bounding boxes, color mapping, and decision-making logic for traffic signals.

Real-time Traffic Signs Detection | *Python, TensorFlow, Keras, OpenCV, scikit-learn*

- Built and trained a CNN model to classify 43 classes of traffic signs with a Softmax-activated output layer.
- Saved the trained model as `.h5` and integrated it with OpenCV for real-time classification from video streams.

Travel Planner Chatbot | *Python*, *Transformers*, *HuggingFace*, *FAISS*, *Streamlit*, *SpeechRecognition*

- Developed an AI-powered chatbot for travel planning using Retrieval-Augmented Generation (RAG), combining speech-to-text, semantic search, and large language models.
- Integrated **Qwen2.5 LLM** fine-tuned on the OSU TravelPlanner dataset with a **FAISS** vector database for efficient information retrieval.
- Implemented voice input with SpeechRecognition API and built an interactive Streamlit UI for real-time queries.
- Enabled the chatbot to generate multi-day itineraries with explanations (justifications) for each recommendation.

EmpathyBot: Sentiment-Driven Chat | *Python*, *HuggingFace Transformers*, *FAISS*, *Streamlit*, *TweetEval*

- Built a sentiment-aware chatbot that classifies user emotions into six categories (joy, sadness, anger, love, surprise, fear) using a pretrained transformer (TweetEval).
- Integrated RAG to retrieve semantically similar tweets with SentenceTransformers and FAISS, enriching responses with context.
- Applied few-shot prompting by augmenting model inputs with (tweet, label) pairs to improve explainability.
- Created a Streamlit web UI displaying emotion prediction, retrieved examples, and chatbot explanations; deployable via Colab/ngrok.

Smart License Plate Detection with OCR | *Python*, *YOLOv8*, *Haar Cascade*, *OpenCV*, *EasyOCR*, *PyQt5*

- Developed a hybrid system for Arabic license plate detection using YOLOv8 and Haar Cascade.
- Applied image preprocessing and EasyOCR to recognize Arabic text and convert it to Latin equivalents.
- Integrated edge detection techniques (Canny, Sobel) to improve visualization and detection quality.
- Built a PyQt5 GUI for image upload and real-time detection using YOLO or Haar Cascade.

Vehicle License Plate & Speed Detection System | *Python*, *YOLOv8*, *OpenCV*, *DeepSORT*, *OpenAI Vision*

- Developed a traffic monitoring system using YOLOv8 and DeepSORT for real-time vehicle detection and tracking.
- Integrated OpenAI Vision to detect and extract license plate information from tracked vehicles.
- Estimated vehicle speeds and displayed tracking IDs, license plates, and speed on processed video frames.
- Generated annotated output videos for traffic monitoring and vehicle analytics.

Bitcoin Price Prediction Web Application | *Python*, *PyTorch*, *LSTM*, *GRU*, *Flask*, *Time-Series Forecasting*

- Developed LSTM and GRU models to predict Bitcoin prices across multiple forecasting horizons (7, 15, 30, 60, and 90 days).
- Implemented a complete data preprocessing pipeline including scaling, sliding window generation, and inverse transformation of predictions.
- Built a Flask web application and REST API to deliver real-time predictions and price statistics.
- Designed a modular architecture with automatic model loading for scalable and reliable deployment.

Bioinformatics Cancer DNA Classification | *Python*, *Machine Learning*, *Deep Learning*, *Bioinformatics*

- Built ML/DL models to classify cancer types from DNA sequence data using bioinformatics techniques.
- Processed genomic data through encoding, feature extraction, and dimensionality reduction.
- Compared multiple models and optimized performance using classical ML and neural networks.
- Evaluated models using accuracy, precision, recall, and ROC-AUC for robust classification.

Gender Identification System | *Python*, *Deep Learning*, *CNN*, *GMM*, *OpenCV*

- Built a CNN-based system for gender classification from facial images.
- Applied preprocessing techniques including resizing, normalization, and data augmentation.
- Combined CNN features with GMM for improved classification performance.
- Integrated OpenCV for face detection and real-time image/video inference.

The Guardian App | *Flutter*, *Supabase*, *FAST API*, *UI/UX*

- Developed a cross-platform mobile application using Flutter to aggregate and display real-time news from multiple categories and sources.
- Integrated FAST APIs to fetch live news data with efficient parsing and dynamic content rendering.
- Implemented Supabase authentication and user management for personalized experience and secure login.
- Designed an intuitive and responsive UI/UX supporting smooth navigation and category-based filtering.
- Optimized app performance with caching strategies and efficient state management.
- Deployed and tested the application across multiple devices ensuring stable performance and production readiness.

Graduation Project 2025 – 2026

The Guardian – AI-Powered Smart Home Security Robot | *YOLOv8, RL, Flutter, Supabase, Raspberry Pi 5, IoT, Zero-Trust*

- Engineered an autonomous mobile security robot integrating Computer Vision, Reinforcement Learning, and Zero-Trust Cybersecurity for active home surveillance.
- Deployed custom **YOLOv8** models combined with **OpenCV** on edge hardware to achieve low-latency human detection, face recognition, and weapon identification.
- Trained a **PPO-based** (Proximal Policy Optimization) Reinforcement Learning agent to enable autonomous navigation, dynamic obstacle avoidance, and optimized patrol routing.
- Developed a cross-platform **Flutter** mobile application featuring live monitoring, instant alert notifications, and remote manual robot override/control.
- Integrated **Supabase** for centralized user authentication, Row-Level Security (RLS), and real-time database synchronization for instant event logging.
- Architected an end-to-end **Zero Trust Architecture** utilizing secure API authentication, encrypted WebSocket communication, and SHA-256 data integrity verification.
- Optimized hardware integration on a **Raspberry Pi 5**, interfacing with camera modules and ultrasonic sensors to process multi-threaded edge-AI surveillance tasks.

Technical Skills

Programming Languages: Python, Java , Dart , C#
Software: Dart , Flutter , PyQt5 , tinkerkad , Flask , Java-GUI
AI/ML& python-libraries:CNN , tensorflow , pytorch , open-cv , sklearn , scipy , MediaPipe , scikit-learn , numpy , pandas , ultralytics(yolo)
Concepts:Artificial Intelligence, Machine Learning, Neural Networks, DeepLearning , NLP , Data Science
Operating Systems:: Windows, Linux (Ubuntu)

Training

Amit January 2024 – August 2024
Machine Learning & AI Diploma *Alexandria, Egypt*

New Horizons September 2023 – January 2024
Flutter Application Development *Alexandria, Egypt*

European Educational Group July 2024 – September 2024
Deep Learning *Alexandria, Egypt*

National Telecommunication Institute(NTI) August 2025 – September 2025
Natural Language Processing *Online*

Languages

Arabic: Mother Tongue
English: Intermediate